THE NUMBER OF ISOLATED VERTICES AT THE TWO ENDS OF THE WINDOW: AT LEAST ONE AND AT MOST D BELOW, NONE ABOVE

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Allowed standard prerequisites (editorial).

- The notation iso(G_p) for the number of isolated vertices of the random subgraph, and the two ends p_- and p_+ of the window together with the parameter omega. Set up in the author's section on the random graph process (source lines 440-520); the expectation n(1-p)^d is computed at the start of the counted proof. Fixes the random variables whose size the statement bounds and the two probabilities at which they are evaluated.

Lemma 0.1. *Whp,*

$$1 \le \mathrm{iso}(G_{p_-}) \le d \qquad \textit{and} \qquad \mathrm{iso}(G_{p_+}) = 0.$$

Proof. Let $Y_p = \mathrm{iso}(G_p)$. Then $\mathbb{E}Y_p = n(1-p)^d$. For $p = p_+$ this expectation is $1/\omega$, and Markov's inequality gives $Y_{p_+} = 0$ whp. For $p = p_-$, we have $\mathbb{E}Y_{p_-} = \omega$, and

$$\mathbb{P}(Y_{p_-} > d) \le \frac{\omega}{d} = o(1).$$

For the lower bound, expand the second moment. Non-adjacent pairs of vertices are simultaneously isolated with probability $(\omega/n)^2$. An adjacent pair of vertices has probability at most $\alpha_M(\omega/n)^2$, where α_M is a suitable constant. Since G has $nd/2$ edges,

$$\mathbb{E}Y_{p_-}^2 \le \omega + \omega^2 + O\left(\frac{d\omega^2}{n}\right) = \omega^2 + O(\omega).$$

Hence $\mathrm{Var}(Y_{p_-}) = O(\omega) = o(\omega^2)$, and Chebyshev's inequality gives $Y_{p_-} > 0$ whp. □

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